Differential Analysis of Gene Selection in CRISPR-Cas9 Screens with MAGeCK

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1 Introduction

CRISPR-Cas9 screening enables genome-wide functional analysis by systematically disrupting genes to study their roles in processes like drug response and cell viability. Pooled CRISPR knockout screens use guide RNAs (gRNAs) to target genes across large sample sets, offering high-throughput insights into gene function. In this study, we analyze a pooled CRISPR-Cas9 dataset comparing control and treatment groups at two timepoints: baseline (T0) and post-treatment (T12), each with five biological replicates. gRNA read counts serve as a proxy for gene selection pressure over time. We applied the MAGeCK framework, which uses robust rank aggregation to estimate gene-level effects from sgRNA data. This allowed us to identify genes under positive or negative selection—those whose knockout enhances or impairs cell survival in response to treatment.

2 Objectives

The primary objective of this study is to identify genes whose knockout significantly influences cell in response to treatment over time. Specifically, we aim to:

- Detect positively selected genes.
- Detect negatively selected genes.
- Compare gene selection dynamics across timepoints and conditions.

3 Methodology

3.1 Data Preparation

The raw CRISPR screening dataset was imported and processed using R. This dataset contained short-guide RNA (sgRNA) identifiers, gene identifiers, and associated sequencing read counts from experimental samples. The primary goal of the analysis was to identify essential genes by comparing treatment conditions with control samples. For downstream analysis, the data was formatted into a standardized MAGeCK-compatible count table, with sgRNA and gene identifiers placed in the first two columns followed by sample-specific count columns. The resulting count table was then exported as a tab-separated file for subsequent analysis.

3.2 Design Matrix Construction

A detailed design matrix was constructed to clearly define the experimental conditions, such as treatment versus control and different timepoints. The assignment of samples to these conditions was based explicitly on naming conventions within the sample identifiers. Samples indicating treatment and specific timepoints were accordingly coded, while control samples were assigned baseline codes. This design matrix allowed robust and clear comparisons across the defined experimental conditions and was also exported as a tab-separated file for use with MAGeCK.

3.3 MAGeCK Analysis

MAGeCK (Model-based Analysis of Genome-wide CRISPR-Cas9 Knockout) employs statistical modeling to identify essential genes by evaluating sgRNA-level abundance changes across experimental conditions. The approach begins with median normalization to adjust for sequencing depth and library size differences among samples. Subsequently, MAGeCK applies a negative binomial model to handle the common overdispersion and variability observed in sgRNA counts. The negative binomial model used by MAGeCK can be mathematically represented as:

$$\mathrm{Var}(X) = \mu + k\mu^b$$

where X represents sgRNA counts, μ is the mean sgRNA count, k is the dispersion parameter, and b is the dispersion exponent. This modeling accounts for variability and differences in sgRNA knockout efficiencies.

MAGeCK ranks sgRNAs based on statistical significance obtained from this model, and uses Robust Rank Aggregation (RRA) to integrate sgRNA-level results to gene-level significance. The RRA method effectively aggregates rankings to highlight genes consistently targeted by significantly enriched or depleted sgRNAs. Permutation testing (with 100 iterations) was applied to enhance robustness and reduce false positives, resulting in adjusted p-values and false discovery rates (FDR). Computational processing was parallelized across 14 threads for efficiency. Ultimately, MAGeCK analysis identified essential genes, providing valuable biological insights for further investigation.

4 Results

4.1 Exploratory Analysis

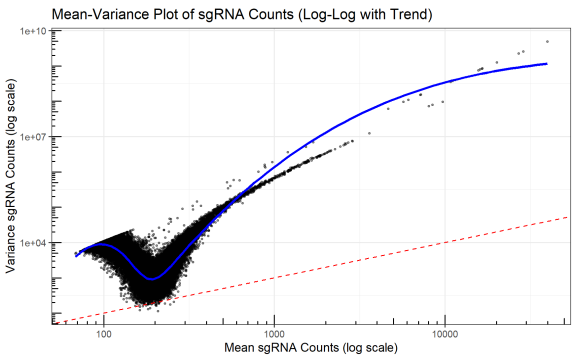

Figure 1: Mean-Variance Plot of sgRNA Counts

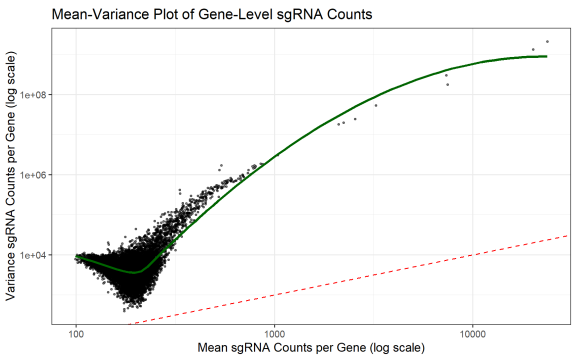

Figure 2: Mean-Variance Plot of Gene-Level sgRNA Counts

Mean-variance plots (log-log scale) were generated at both the sgRNA and gene levels to evaluate overdispersion in the CRISPR screen data. The sgRNA-level plot revealed that variance consistently exceeded the mean, especially at higher expression levels, with a distinct "V" shape at low counts due to technical noise. The gene-level plot, which averaged guide-level data per gene, showed a smoother trend but similarly highlighted overdispersion. These observations confirmed that the data deviated significantly from Poisson (red

dashed line) expectations, supporting the use of MAGeCK's negative binomial model to appropriately account for variability across samples and guides.

4.2 Differential Analysis Results

4.2.1 WALD FDR vs Standard FDR

Regulation	Gene Count	Method
Not significant	9,995	Standard FDR
Positive	5	Standard FDR
Negative	4,450	Wald FDR
Not significant	3,065	Wald FDR
Positive	2,485	Wald FDR

Table 1: Gene Regulation Summary by Method

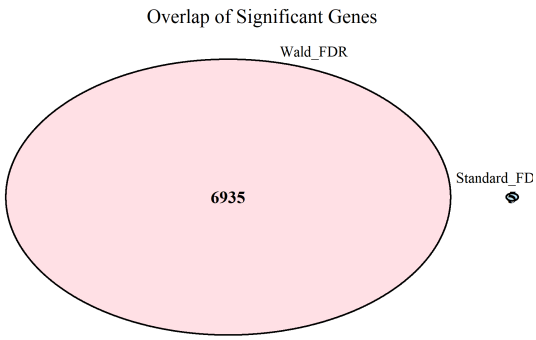

Figure 3: Significant Genes (WALD vs Standard)

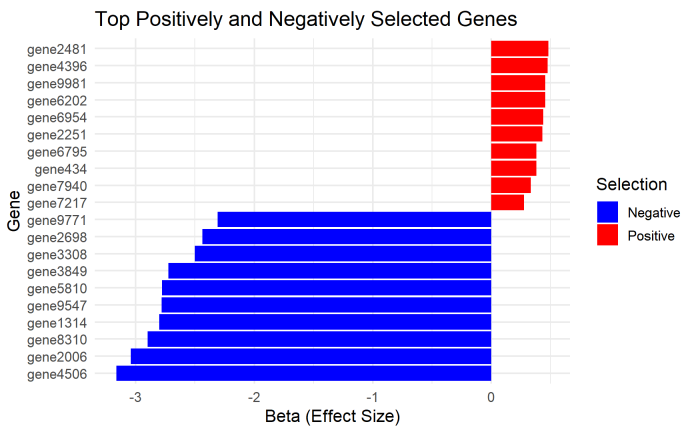

Figure 4: Top Significant Genes(WALD)

In our pooled CRISPR-Cas9 knockout screen analysis using the MAGeCK MLE model, we evaluated gene-level significance based on both the Wald test and the standard FDR correction. The Wald test, integral to the MLE framework, estimates gene effects through regression modeling and provides two key metrics: the Wald p-value and Wald FDR. These are specifically designed to account for experimental design factors such as timepoint and treatment. In contrast, the standard FDR represents a more generic multiple testing correction and is not tailored to the underlying regression model.

When comparing results, the Wald FDR identified 6,935 genes as significant (FDR $<$ 0.05), while the standard FDR identified only 5. Notably, there was no overlap between the two sets, as confirmed by Venn diagram analysis. This sharp contrast indicates that the standard FDR is overly conservative and may underestimate biological signals in model-based analyses. Therefore, following best practices and the MAGeCK documentation, we focused our downstream interpretation and visualization on Wald FDR

values, using them to define significantly selected genes under positive ($\beta > 0$) or negative ($\beta < 0$) selection pressure.

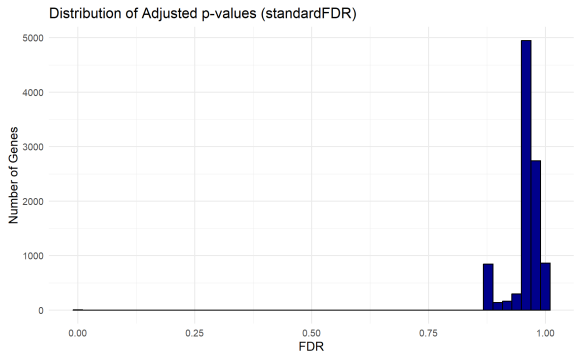

Figure 5: Standard FDR vs Number of genes

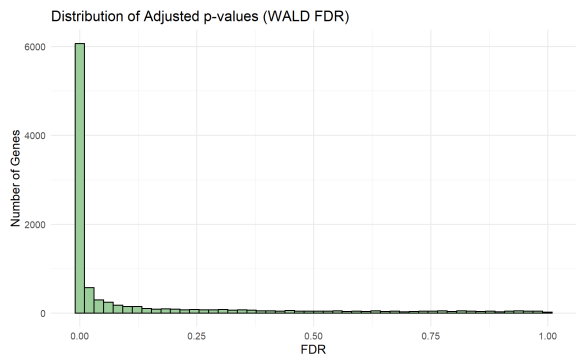

Figure 6: WALD FDR vs Number of genes

Histogram of Wald p-values illustrates a strong enrichment of genes with low p-values, indicating potential biological signal. In contrast, standard p-values skewed toward 1, consistent with a lack of enrichment or reduced sensitivity.

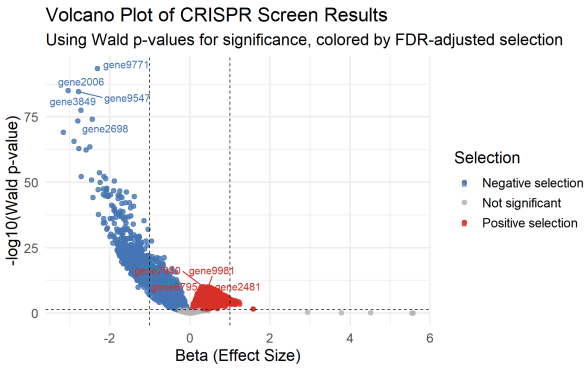

Figure 7: Volcano plot

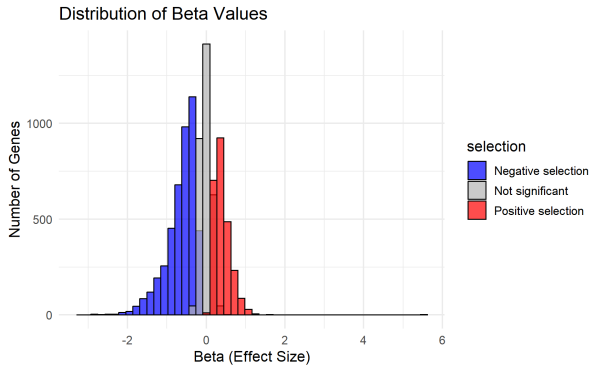

Figure 8: Density plot

Data was visualized to explore gene-level effects in the CRISPR screen using a volcano plot and a histogram of Beta (effect size) values. The volcano plot integrates statistical significance (–log10 Wald p-value) with effect size, highlighting genes under strong selection. Genes under significant negative selection (blue) exhibit large negative Beta values, suggesting that their knockout impairs cell viability—marking them as potentially essential. In contrast, positively selected genes (red) show positive Beta values, indicating that their loss may confer a growth or survival advantage, possibly in response to treatment. Key negatively selected genes such as gene9771 and gene2006 were among the most statistically significant.

Complementing this, the histogram of Beta values reveals the broader distribution of gene effects across the dataset. Most genes are centered around a Beta of zero, reflecting neutrality, while smaller subsets display distinct positive or negative shifts, corresponding to selection pressure. This distribution supports the volcano plot findings, confirming that while most genes do not exert strong effects, a clear subset significantly influences cellular fitness under the experimental conditions.

5 Conclusion

In conclusion, we conducted a differential analysis of a pooled CRISPR-Cas9 knockout screen by comparing control and treatment groups at two different time points using the MAGeCK MLE method. Our results identified several genes that were strongly negatively selected, highlighting their essential role in cell viability. Additionally, we discovered a few positively selected genes that may be linked to treatment resistance or adaptation. The volcano plot and FDR-based comparisons revealed significant differences between the Wald method and the standard FDR approach, with the Wald method identifying a larger number of significant genes. Overall, this analysis emphasizes key candidates for further investigation as potential therapeutic targets or markers of drug response, offering valuable insights into gene dependencies under treatment pressure.

A R and Linux Codes

```r
# Primary handling data with R:

library(dplyr)
library(ggplot2)

# CRISPR dataset
data <- read.delim("CRISPR_dataset.txt", stringsAsFactors = FALSE)

# Create MAGeCK count table:
# - 1st column = sgRNA (guideNames)
# - 2nd column = Gene (geneNames)
# - Followed by sample count columns

mageck_counts <- data %>%
  select(sgRNA = guideNames, Gene = geneNames, everything())

write.table(mageck_counts, "mageck_counts.txt", sep = "\t", quote = FALSE, row.names = FALSE)

# Extract sample names from column names (excluding sgRNA and Gene)
samples <- colnames(mageck_counts)[-(1:2)]

# Create design matrix (samples + experimental conditions)
design_matrix <- data.frame(
  sample = samples,
  treatment = ifelse(grepl("treat", samples), 1, 0),
  timepoint = ifelse(grepl("t12", samples, ignore.case = TRUE), 1, 0)
)

write.table(design_matrix, "design_matrix.txt", sep = "\t", quote = FALSE, row.names = FALSE)

```

```r

count_matrix <- mageck_counts[,-c(1,2)]

mean_counts <- rowMeans(count_matrix)
variance_counts <- apply(count_matrix, 1, var)

df <- data.frame(Mean = mean_counts, Variance = variance_counts)

# Plot mean-variance plot:

ggplot(df, aes(x = Mean, y = Variance)) +
  geom_point(alpha = 0.4, size = 0.6) +
  scale_x_log10() +
  scale_y_log10() +
  geom_smooth(method = "loess", se = FALSE, color = "blue", linewidth = 1) +
  geom_abline(intercept = 0, slope = 1, color = "red", linetype = "dashed") +
  annotation_logticks(sides = "lb") +
  labs(title = "Mean-Variance Plot of sgRNA Counts (Log-Log with Trend)",
       x = "Mean sgRNA Counts (log scale)",
       y = "Variance sgRNA Counts (log scale)") +
  theme_bw()


# Merge back gene info
df$Gene <- mageck_counts$Gene

# Aggregate by gene
df_gene <- df %>%
  group_by(Gene) %>%
  summarise(Mean = mean(Mean), Variance = mean(Variance))


ggplot(df_gene_filtered, aes(x = Mean, y = Variance)) +
  geom_point(alpha = 0.5, size = 0.7) +
  scale_x_log10() +
  scale_y_log10() +
  geom_smooth(method = "loess", se = FALSE, color = "darkgreen") +
  geom_abline(intercept = 0, slope = 1, color = "red", linetype = "dashed") +
  labs(title = "Mean-Variance Plot of Gene-Level sgRNA Counts",
       x = "Mean sgRNA Counts per Gene (log scale)",
       y = "Variance sgRNA Counts per Gene (log scale)") +
  theme_bw()

-------------------

# Handing to Linux
```

```

mageck mle \
  --count-table mageck_counts.txt \
  --design-matrix design_matrix.txt \
  --norm-method median \
  --permutation 100 \
  --threads 14 \
  --output-prefix mageck_output_final

-------------------

# Back to R for visualisation


# Load all required libraries once
library(ggplot2)
library(dplyr)
library(ggrepel)
library(VennDiagram)
library(grid)

# Load MAGeCK gene summary data
df <- read.table("mageck_output_final.gene_summary.txt",
                 header = TRUE,
                 sep = "\t",
                 check.names = FALSE,
                 comment.char = "")

# Rename and enrich dataframe
df <- df %>%
  rename(
    gene = Gene,
    beta = `timepoint|beta`,
    pval = `timepoint|wald-p-value`,
    fdr = `timepoint|fdr`,
    wald_fdr = `timepoint|wald-fdr`
  ) %>%
  mutate(
    log10_pvalue = -log10(pval),
    log10_fdr = -log10(wald_fdr),
    significance = case_when(
      wald_fdr < 0.05 & beta > 0 ~ "Positive selection",
      wald_fdr < 0.05 & beta < 0 ~ "Negative selection",
      TRUE ~ "Not significant"
    )
  )


top_labels <- df %>%
  filter(wald_fdr < 0.01) %>%
  arrange(wald_fdr) %>%
```

```r
  slice_head(n = 10)

ggplot(df, aes(x = beta, y = log10_pvalue, color = significance)) +
  geom_point(alpha = 0.8, size = 1.8) +
  geom_text_repel(data = top_labels, aes(label = gene), size = 3.2, max.overlaps = 15) +
  scale_color_manual(values = c(
    "Positive selection" = "#d73027",
    "Negative selection" = "#4575b4",
    "Not significant" = "gray70"
  )) +
  geom_vline(xintercept = c(-1, 1), linetype = "dashed", color = "black") +
  geom_hline(yintercept = -log10(0.05), linetype = "dashed", color = "black") +
  labs(
    title = "Volcano Plot of CRISPR Screen Results",
    subtitle = "Using Wald p-values",
    x = "Beta (Effect Size)",
    y = "-log10(Wald p-value)",
    color = "Selection"
  ) +
  theme_minimal(base_size = 14)


# Histogram: Standard FDR
ggplot(df, aes(x = fdr)) +
  geom_histogram(bins = 50, fill = "tomato", color = "black") +
  labs(title = "Distribution of Standard FDR", x = "Standard FDR", y = "Count") +
  theme_minimal()

# Histogram: Wald FDR
ggplot(df, aes(x = wald_fdr)) +
  geom_histogram(bins = 50, fill = "darkseagreen3", color = "black") +
  labs(title = "Distribution of Wald FDR", x = "Wald FDR", y = "Count") +
  theme_minimal()


wald_sig_genes <- df$gene[which(df$wald_fdr < 0.05)]
std_sig_genes  <- df$gene[which(df$fdr < 0.05)]

venn.plot <- venn.diagram(
  x = list(
    Wald_FDR = wald_sig_genes,
    Standard_FDR = std_sig_genes
  ),
  filename = NULL,
```

```
  fill = c("lightpink", "skyblue"),
  alpha = 0.6,
  cex = 2,
  cat.cex = 1.5,
  fontface = "bold",
  cat.fontface = "bold",
  main = "Overlap of Significant Genes (FDR < 0.05)",
  main.cex = 2
)

grid.newpage()
grid.draw(venn.plot)


# Summary from both FDR methods
summary_combined <- bind_rows(
  df %>%
    mutate(method = "Standard FDR") %>%
    mutate(regulation = case_when(
      fdr < 0.05 & beta > 0 ~ "Positive",
      fdr < 0.05 & beta < 0 ~ "Negative",
      TRUE ~ "Not significant"
    )) %>%
    count(regulation, method),

  df %>%
    mutate(method = "Wald FDR") %>%
    mutate(regulation = case_when(
      wald_fdr < 0.05 & beta > 0 ~ "Positive",
      wald_fdr < 0.05 & beta < 0 ~ "Negative",
      TRUE ~ "Not significant"
    )) %>%
    count(regulation, method)
)

ggplot(summary_combined, aes(x = method, y = n, fill =
   regulation)) +
  geom_col(position = "stack") +
  labs(title = "Comparison of Gene Selection by Method",
       y = "Number of Genes",
       x = "FDR Method",
       fill = "Selection") +
  theme_minimal()


top_pos <- df %>%
  filter(significance == "Positive selection") %>%
  arrange(wald_fdr) %>%
  slice_head(n = 10)

top_neg <- df %>%
```

```r
  filter(significance == "Negative selection") %>%
  arrange(wald_fdr) %>%
  slice_head(n = 10)

top_combined <- bind_rows(top_pos, top_neg)

ggplot(top_combined, aes(x = reorder(gene, beta), y = beta, fill
    = significance)) +
  geom_col() +
  coord_flip() +
  scale_fill_manual(values = c("Positive selection" = "#d73027",
      "Negative selection" = "#4575b4")) +
  labs(
    title = "Top Positively and Negatively Selected Genes",
    x = "Gene",
    y = "Beta (Effect Size)",
    fill = "Selection"
  ) +
  theme_minimal(base_size = 13)


standard_summary <- df %>%
  mutate(
    regulation = case_when(
      fdr < 0.05 & beta > 0 ~ "Positive",
      fdr < 0.05 & beta < 0 ~ "Negative",
      TRUE ~ "Not significant"
    )
  ) %>%
  group_by(regulation) %>%
  summarise(gene_count = n()) %>%
  mutate(method = "Standard FDR")

# Categorize using Wald FDR
wald_summary <- df %>%
  mutate(
    regulation = case_when(
      wald_fdr < 0.05 & beta > 0 ~ "Positive",
      wald_fdr < 0.05 & beta < 0 ~ "Negative",
      TRUE ~ "Not significant"
    )
  ) %>%
  group_by(regulation) %>%
  summarise(gene_count = n()) %>%
  mutate(method = "Wald FDR")

# Combine both summaries
summary_combined <- bind_rows(standard_summary, wald_summary)

# Print combined summary
print(summary_combined)
```

```r

# Define positively and negatively selected gene subsets
positive_genes <- df %>%
  filter(wald_fdr < 0.05, beta > 0)

negative_genes <- df %>%
  filter(wald_fdr < 0.05, beta < 0)


ggplot(positive_genes, aes(x = beta)) +
  geom_histogram(binwidth = 0.2, fill = "steelblue", color =
     "black") +
  labs(
    title = "Distribution of Positively Selected Genes",
    x = "Log2 Fold Change",
    y = "Number of Genes"
  ) +
  theme_minimal()

# Plot distribution of negative selection
ggplot(negative_genes, aes(x = beta)) +
  geom_histogram(binwidth = 0.2, fill = "tomato", color =
     "black") +
  labs(
    title = "Distribution of Negatively Selected Genes",
    x = "Log2 Fold Change",
    y = "Number of Genes"
  ) +
  theme_minimal()
```